

Weighted corrector limits and an actual martingale estimate

HOCF–RIESZ, Rounds 011–012

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Scope and audit record. This internal synthesis records the exact THM–032 and THM–033 implications. Complete proofs, independent reconstructions and hostile dispositions are preserved separately in the campaign memoranda and audit ledgers. Earlier particle and full-inverse modules are used only in their stated scopes. Neither module proves the cubic drift residual or a fluctuation law.

The full inverse at bounded rescaled diffusivity

Use the unit Haar torus, coefficient-one periodic Riesz kernel, $K = -\nabla g$, zero external drift, and a smooth real terminal test h . Fix

$$d \geq 3, \quad 0 < s \leq d - 2, \quad 0 \leq T, L < \infty, \quad N \geq 2, \quad 0 \leq \nu \leq LN^{-2/p}, \quad p = s + 2.$$

The actual homogeneous Fourier test and source are

$$f_t^\nu = \widehat{h}(0) + \sum_{k \neq 0} \widehat{h}(k) e^{-(T-t)(4\pi^2\nu|k|^2 + d_k)} e^{2\pi i k \cdot x}, \quad d_k = 4\pi^2 c_{d,s} |k|^{s+2-d},$$

$$J_t^\nu(x, y) = K(x - y) \cdot (\nabla f_t^\nu(x) - \nabla f_t^\nu(y)).$$

Let $S^{N,\nu}$ be the genuine auxiliary pair evolution and let $D = \operatorname{div} K$ be the complete compensated finite signed measure. Both response slots remain:

$$\mathcal{R}v = - \int v(x + w, y) D(dw) - \int v(x, y + w) D(dw),$$

$$\Phi_t^{N,\nu} = \int_t^T S_{a-t}^{N,\nu} (J_a^\nu + \mathcal{R}\Phi_a^{N,\nu}) da.$$

For the fixed positive periodic weights $w_q = w_1^q$, equal to $|x - y|^{-q}$ near the diagonal, define

$$\|v\|_X = \sup_{x \neq y} \frac{|v|}{w_s} + \sup_{x \neq y} \frac{|\nabla_{x,y} v|}{w_{s+1}}.$$

Then $\sup_{N,\nu,t} \|\Phi_t^{N,\nu}\|_X \leq C$. The derivatives are actual jointly continuous derivatives off the diagonal and global weak Haar derivatives. In particular the kernels are uniformly bounded in $W^{1,1}$, and the one-variable contractions and their first derivatives are continuous and uniformly bounded.

The mechanism is explicit. For every moment order $\alpha > m$, the radial weighted generator has the retained negative term $-2s(\alpha - m)r^{-p}/N$ and possible positive term $2\nu\alpha(\alpha + 2 - d)_+r^{-2}$. Maximizing half of the latter competition costs a fixed multiple of

$$N^{2/s} \nu^{p/s} = (\nu N^{2/p})^{p/s} \leq L^{p/s}.$$

The fixed annular costs are also uniform. Terminal moments control the source in its own value and gradient weights. Both responses are bounded on X , and the full ordered Volterra series has a uniform factorial tail. Common-start flow construction and higher-moment uniform integrability justify the expectation derivatives before these uniform estimates are used.

Identification and topology of the limit

The limiting full kernel retains the response operator:

$$\Phi_t^0 = \int_t^T e^{(a-t)\mathcal{R}} J_a^0 da, \quad \partial_t \Phi_t^0 + \mathcal{R}\Phi_t^0 = -J_t^0, \quad \Phi_T^0 = 0.$$

The exponential is the norm-convergent series of the bounded signed operator on X and on the pointwise weighted-value space Y_s . The source is continuous into X , so the integral is a genuine norm-valued integral. Factorial iteration gives uniqueness among uniformly Y_s -bounded pointwise mild solutions.

For every compact H away from the diagonal,

$$\sup_{0 \leq \nu \leq LN^{-2/p}} \sup_{t \leq T} \|\Phi_t^{N,\nu} - \Phi_t^0\|_{C^1(H)} \longrightarrow 0,$$

$$\sup_{0 \leq \nu \leq LN^{-2/p}} \sup_{t \leq T} \|\Phi_t^{N,\nu} - \Phi_t^0\|_{W^{1,1}(\text{Haar}^2)} \longrightarrow 0.$$

On a fixed compact set, small Brownian motion, drift of order $1/N$ and its Jacobian give the base evolution's local convergence to the identity. Higher moments control the complementary event. Separated-singularity bounds preserve this local convergence under both nonlocal responses. First compare finitely many Volterra terms; then use their uniform tails. Finally delete a relative tube of radius δ . Its value and gradient integrals are bounded by

$$C\delta^{d-s} \quad \text{and} \quad C\delta^{d-s-1},$$

respectively. Sending N to infinity before δ to zero gives the global result and uniform contraction convergence. No uncountable intersection of probability-one sets over diffusivities is needed.

The limit need not be bounded at the diagonal, and its gradient need not belong to Haar L^2 . There is no assertion of weighted-supremum, second-derivative or time-derivative convergence. These deterministic kernel estimates alone do not transfer to a correlated particle law.

A separate, stronger derivative weight

For the actual-noise implication restrict to

$$0 < s < d - 2, \quad a = s/p, \quad \theta = 1 - s/d, \quad \nu N^{2/p} \leq L, \quad 0 \leq \nu \leq \nu_*.$$

For each $1 < q < d/2$ with $q \leq s + 1$, a separate argument proves

$$\sup_t \sup_{x \neq y} \frac{|\nabla_{x,y} \Phi_t|}{w_q} \leq C_q N^{(s+1-q)/p}.$$

Set $\eta = s + 1 - q$. The weighted stopped evolution retains both a terminal bound and an occupation bound with coefficient $1/N$. Split the actual source-gradient weight at the repulsion scale:

$$w_{s+1} \leq CN^{\eta/p} w_q + N^{\eta/p-1} \mathbf{1}_{\{r \leq 1/16\}} w_{q+p}.$$

The occupation factor N cancels the last coefficient's N^{-1} . The derivative-only homogeneous response bound has no extra value-amplitude term, so Gronwall on the genuine full inverse gives the stated power. The separate THM-032 spatial limit is not a prerequisite for this proof.

The genuine interacting-law bracket

The actual particles start iid Haar independently of their Brownian drivers:

$$dX_i = B_i dt + \sqrt{2\nu} dW_i, \quad B_i = \frac{1}{N} \sum_{j \neq i} K(X_i - X_j), \quad H_N = \frac{1}{N} \sum_{i < j} g(X_i - X_j).$$

At positive noise put $\beta = 1/\nu$, $b_N = \min(\beta, 1)$ and $\sigma_N^2 = Nb_N$. The exact earlier energy identity and sharp R10 floor give

$$\mathbb{E}H_N(X_T) + \mathbb{E} \int_0^T \sum_i |B_i|^2 dt + \nu(N-1) \int_0^T \mathbb{E}D_{\text{cl}}(X_1 - X_2) dt = 0, \quad H_N \geq -CN^{s/d}.$$

Strictly below Coulomb, $D_{\text{cl}} \geq cw_p - C$ with $c > 0$ because $s(d-2-s) > 0$. Discarding only the whole nonnegative force square therefore proves

$$\nu \mathbb{E} \int_0^T w_p(X_1 - X_2) dt \leq C(N^{-\theta} + \nu).$$

No individual pair-force square is extracted from a signed expansion. At Coulomb the positive singularity density coefficient vanishes, so this occupation argument is unavailable.

Choose $q = p/2 \in (1, d/2)$. The gradient estimate gives $|G|^2 \leq CN^a w_p$, with $G = \nabla_x \Phi$ and $A(x) = \int G(x, y) dy$. The exact ordered deleted-pair statistic satisfies

$$\nabla_i P_N[\Phi] = N^{-2} \sum_{j \neq i} G(X_i, X_j) - N^{-1} A(X_i).$$

Exchangeability, the actual one-body Haar law and the finite-sum square inequality yield

$$Q_N := 2\nu Nb_N \mathbb{E} \int_0^T \sum_i |\nabla_i P_N[\Phi_t]|^2 dt \leq 4\nu b_N \int_0^T \left[\frac{(N-1)^2}{N^2} \mathbb{E}|G_{12}|^2 + \|A\|_2^2 \right] dt \leq Cb_N N^a (N^{-\theta} + \nu).$$

This is the entire actual bracket of the genuine earlier-domain martingale. It is not a Haar reference integral or a clipped replacement. The leading scaled bracket is bounded by $C\nu b_N \leq C$; the expected time integral of absolute scaled cross-variation is at most $C\sqrt{Q_N}$. Zero-noise brackets vanish directly, without dividing by zero.

At microscopic criticality $\lambda_N = \beta_N N^{-\theta} \rightarrow \lambda > 0$,

$$\nu_N = \lambda_N^{-1} N^{-\theta}, \quad \nu_N N^{2/p} = \lambda_N^{-1} N^{s(p-d)/(dp)} \rightarrow 0.$$

Hence $Q_N \rightarrow 0$ whenever $s(s+2) < 2d$. On any bounded-rescaled-diffusivity sequence it also vanishes if $s < \min(2, d-2)$. Equality boundaries and the Coulomb endpoint are excluded. General unrestricted bounded-noise sequences require a separate argument.

Within these noise ranges, the next open load-bearing assertion is the actual evolved cubic drift residual with every lower contraction retained. The broader actual-noise range and critical hierarchy remain open. The old condition $\beta_N N^{2s/d-1} \rightarrow 0$, full subcriticality $\lambda_N \rightarrow 0$, and finite positive criticality remain distinct; the logarithmic case is separate. No completed fluctuation law is asserted.